Ticket 8

TCK08-HostsFile

Topics: hosts file, name resolution order, DNS client cache, short name vs FQDN, UNC paths vs mapped drives

1. Summary

Title	User can't reach file server via FQDN — hosts file poisoned
Date	2026-10-31
Ticket ID	TCK08-HostsFile
Status	Resolved

2. Problem

• **Symptom:** User (acarter) reported being unable to connect to the Sales share — either timing out or connecting to the wrong destination. Internet and Outlook worked fine. (Simulated: a hosts file entry mapping the server's FQDN to a fake IP.)

• **Impact:** Access to the Sales share when reached via the server's FQDN.

• **Started when:** After the break script was administered on the client VM.

• **Recent changes:** An entry was added to the client's hosts file mapping `WIN-4G8SQK9BMC1.acme.local` to 192.168.50.200 (a non-existent IP).

3. Environment

• **Device / OS:** Client VM, Windows, joined to acme.local

• **Network / Server:** Domain Controller WIN-4G8SQK9BMC1 (real IP 192.168.50.10), AD lab

• **Software / Version:** PowerShell, Notepad

4. Troubleshooting Steps

#	What I tried	Result
1	Accessed `\\WIN-4G8SQK9BMC1\Sales` (bare name) via Run	Worked — break appeared not to have landed
2	Accessed `\\WIN-4G8SQK9BMC1.acme.local\Sales` (FQDN) via Run	Failed — confirmed the break only affects FQDN lookups
3	Ran `Get-Content C:\Windows\System32\drivers\etc\hosts`	Found the poisoned entry mapping the FQDN to 192.168.50.200
4	Ran `Clear-DnsClientCache`, then re-ran the break script to confirm the hosts edit was actually carried out	Confirmed poisoned entry was present as expected
5	Attempted to edit the hosts file as a standard user	Blocked — requires admin rights (protected system file)
6	Opened Notepad elevated (as Administrator), opened the hosts file, removed the poisoned entry	Entry removed
7	Ran `Clear-DnsClientCache` again	Cache cleared
8	Re-accessed `\\WIN-4G8SQK9BMC1.acme.local\Sales`	Worked — resolved correctly to the real DC

5. Root Cause

The hosts file (`C:\Windows\System32\drivers\etc\hosts`) is checked before any DNS query is made, and matches only the exact string queried. An entry for the server's full FQDN was poisoned with a fake IP, so FQDN lookups failed while bare-name lookups (which never match that entry) resolved normally via DNS.

6. Fix

Removed the poisoned entry from the hosts file via an elevated Notepad session, then cleared the DNS client cache.

```
notepad C:\Windows\System32\drivers\etc\hosts
Clear-DnsClientCache
```

Commands used during troubleshooting:

```
Get-Content C:\Windows\System32\drivers\etc\hosts
Clear-DnsClientCache
notepad C:\Windows\System32\drivers\etc\hosts
```

7. Verification

After removing the entry and clearing the cache, `\\WIN-4G8SQK9BMC1.acme.local\Sales` resolved and connected successfully, confirming the machine could reach the domain controller via DNS with no hosts file interference.

8. Lessons Learned

• **What I'd check first next time:** Check the hosts file before assuming a pure DNS/network issue, especially if the symptom is inconsistent depending on exactly which name format (short vs FQDN) is used.

• **What I got wrong or assumed:** Initially assumed the break hadn't landed because the bare-name path still worked. The hosts file entry was narrow — matched only the literal FQDN string typed, not the short name — so bare-name queries skipped it entirely and went straight to DNS.

• **How to prevent it:** This ticket's design has a real weakness — it only manifests if the user happens to type the FQDN, which isn't realistic for a typical user who'd normally use a mapped drive or short name. A more realistic version would poison both the short name and FQDN, or target whatever name a mapped drive would actually use. Also confirmed that a mapped drive is just a saved shortcut to a UNC path (`\\server\share`) — it still triggers the same name resolution underneath, just hidden from the user, and can sometimes mask a break for longer by reusing a cached connection instead of resolving fresh each time.

9. Screenshots

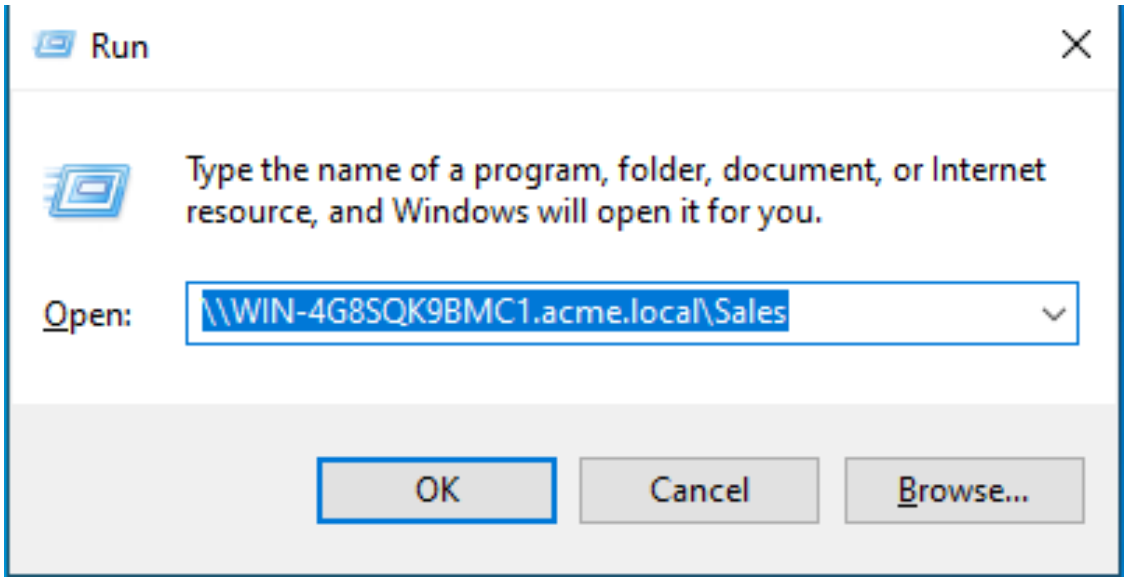


Figure 1. Run dialog with \\WIN-4G8SQK9BMC1.acme.local\Sales

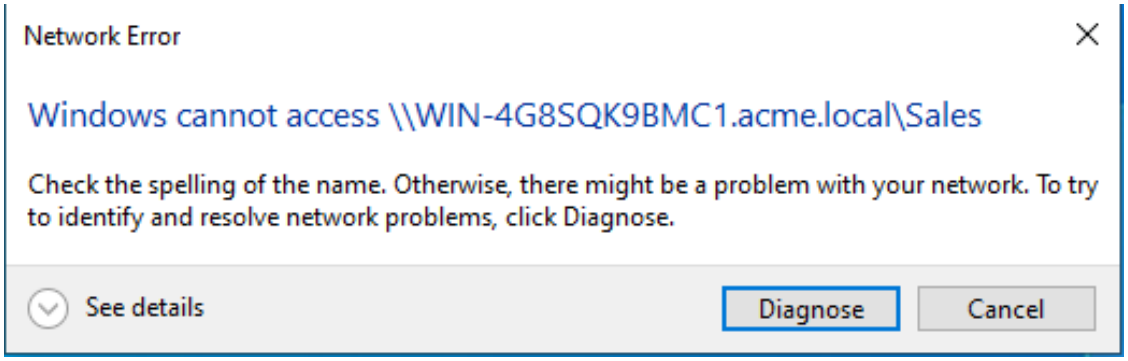


Figure 2. Network Error: Windows cannot access \\WIN-4G8SQK9BMC1.acme.local\Sales

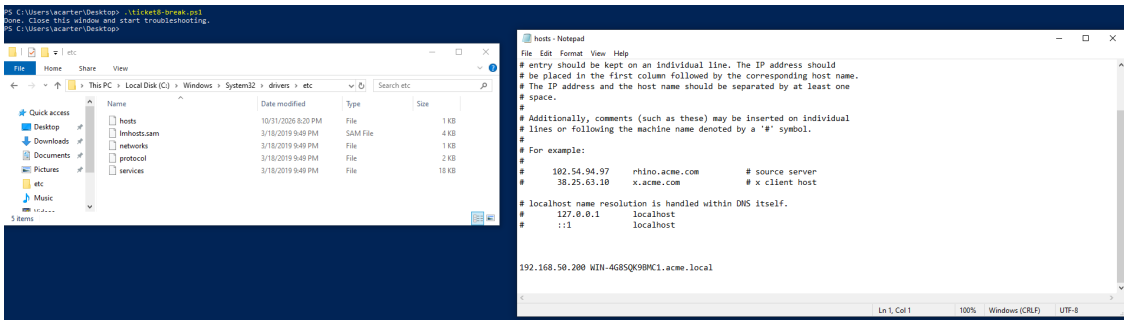


Figure 3. hosts file in Notepad after the break script ran, with the entry mapping WIN-4G8SQK9BMC1.acme.local to 192.168.50.200

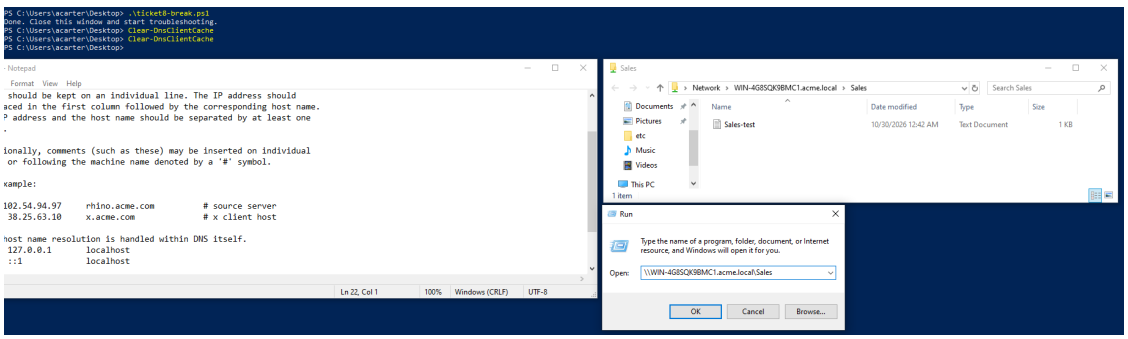


Figure 4. hosts file with the entry removed, DNS cache cleared, and the Sales share opening through the FQDN